Class 2

Colin Kuehl

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Today we'll explore the gapminder data. It contains life expectancy, population, and GDP per capita for countries every five years from 1952 to 2007. The gapminder data comes from the Gapminder Foundation: https://www.gapminder.org/data/ made famous by Hans Rosling's TED talks. We'll continue our exploration using R to understand and visualize data.

Part 1: Getting to Know Gapminder

Make sure you have opened the Rproject first.

Today's data comes packaged inside R itself. Instead of loading a file, we install a package once, then load it with library() each session.

Packages are like bonus material for R. We will use them all the time. Installing puts it on your computer. Library opens it(like a book) so everytime you use it you need library in your code first.

```
#install.packages("gapminder") #only run this once ever - then put the # back in front
library(gapminder)
data("gapminder") #Puts into our environment

#install.packages("dplyr") #We'll use this package later, but always a good idea to load pack
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag
```

```
The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

Looking over the data using commmands from yesterday

(hint: use code from yesterday)

```r
head(gapminder)
```

```
# A tibble: 6 x 6
  country     continent  year lifeExp      pop gdpPercap
  <fct>       <fct>     <int>   <dbl>    <int>     <dbl>
1 Afghanistan Asia       1952    28.8  8425333      779.
2 Afghanistan Asia       1957    30.3  9240934      821.
3 Afghanistan Asia       1962    32.0 10267083      853.
4 Afghanistan Asia       1967    34.0 11537966      836.
5 Afghanistan Asia       1972    36.1 13079460      740.
6 Afghanistan Asia       1977    38.4 14880372      786.
```

```r
tail(gapminder)
```

```
# A tibble: 6 x 6
  country  continent  year lifeExp      pop gdpPercap
  <fct>    <fct>     <int>   <dbl>    <int>     <dbl>
1 Zimbabwe Africa     1982    60.4  7636524      789.
2 Zimbabwe Africa     1987    62.4  9216418      706.
3 Zimbabwe Africa     1992    60.4 10704340      693.
4 Zimbabwe Africa     1997    46.8 11404948      792.
5 Zimbabwe Africa     2002    40.0 11926563      672.
6 Zimbabwe Africa     2007    43.5 12311143      470.
```

```r
summary(gapminder)
```

```
        country        continent        year         lifeExp     
 Afghanistan:  12   Africa  :624   Min.   :1952   Min.   :23.60  
 Albania    :  12   Americas:300   1st Qu.:1966   1st Qu.:48.20  
 Algeria    :  12   Asia    :396   Median :1980   Median :60.71  
 Angola     :  12   Europe  :360   Mean   :1980   Mean   :59.47  
 Argentina  :  12   Oceania : 24   3rd Qu.:1993   3rd Qu.:70.85  
```

```
 Australia  :  12                  Max.   :2007   Max.   :82.60
 (Other)    :1632
      pop              gdpPercap
 Min.   :6.001e+04   Min.   :   241.2
 1st Qu.:2.794e+06   1st Qu.:  1202.1
 Median :7.024e+06   Median :  3531.8
 Mean   :2.960e+07   Mean   :  7215.3
 3rd Qu.:1.959e+07   3rd Qu.:  9325.5
 Max.   :1.319e+09   Max.   :113523.1
```

```
dim(gapminder) #Count of how many observations and variables
```

```
[1] 1704    6
```

```
str(gapminder) #This tells us what kind of data R thinks each variable is - we'll talk about
```

```
tibble [1,704 x 6] (S3: tbl_df/tbl/data.frame)
 $ country  : Factor w/ 142 levels "Afghanistan",..: 1 1 1 1 1 1 1 1 1 1 ...
 $ continent: Factor w/ 5 levels "Africa","Americas",..: 3 3 3 3 3 3 3 3 3 3 ...
 $ year     : int [1:1704] 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 ...
 $ lifeExp  : num [1:1704] 28.8 30.3 32 34 36.1 ...
 $ pop      : int [1:1704] 8425333 9240934 10267083 11537966 13079460 14880372 12881816 1386
 $ gdpPercap: num [1:1704] 779 821 853 836 740 ...
```

```
length(unique(gapminder$country)) #gives a count of unique countrys
```

```
[1] 142
```

```
length(unique(gapminder$year))
```

```
[1] 12
```

Checking in (with a partner)

1) How many countries are in the dataset?
2) What is the average life expectancy?
3) What is the median GDP per capita? Average?

a) Create a scatterplot with the relationship between population and Life Expectancy. Add a trend line b)What country has the highest life expectancy?

```
length(unique(gapminder$country))
```

```
[1] 142
```

```
mean(gapminder$lifeExp)
```

```
[1] 59.47444
```

```
median(gapminder$gdpPercap)
```

```
[1] 3531.847
```

It is best practice to "render as you go." Try rendering now to make sure you don't have errors in your code.

Country and continent are not continuous(numeric in R speak) variables. They are categorical. We can get counts in each category.

frequency tables:

```
table(gapminder$continent) #gives count of different categories
```

```

  Africa Americas     Asia   Europe  Oceania 
     624      300      396      360       24 
```

```
table(gapminder$year)
```

```

1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 
 142  142  142  142  142  142  142  142  142  142  142  142 
```

```
prop.table(table(gapminder$continent)) #gives percentages
```

```
    Africa   Americas       Asia     Europe    Oceania 
0.36619718 0.17605634 0.23239437 0.21126761 0.01408451 
```

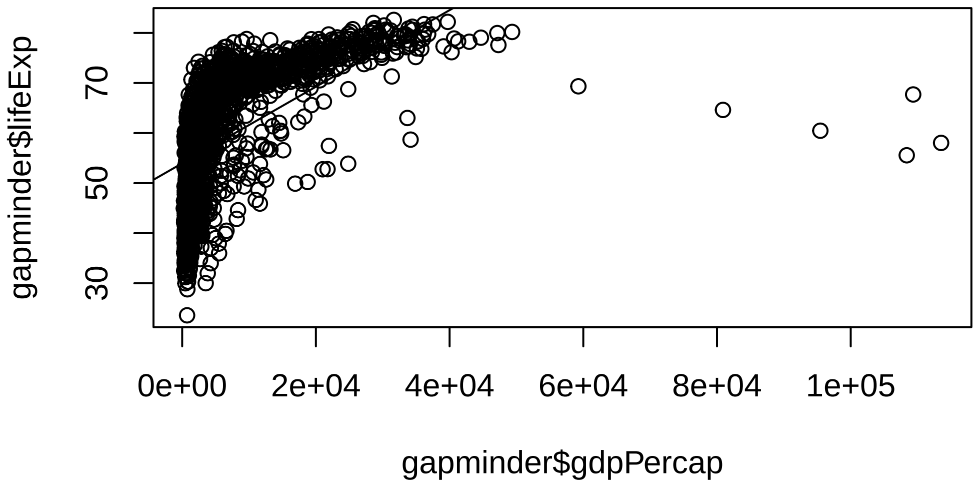
I use this command all the time to get a sense of the data.

Check in: What percentage of the observations in the dataset are from Asia?

“

Plot the relationship between GDP per capita and life expectancy

```
plot(gapminder$gdpPercap, gapminder$lifeExp) #scatterplot (X then Y)
abline(lm(gapminder$lifeExp~gapminder$gdpPercap)) #add trend line (Y then X)
```



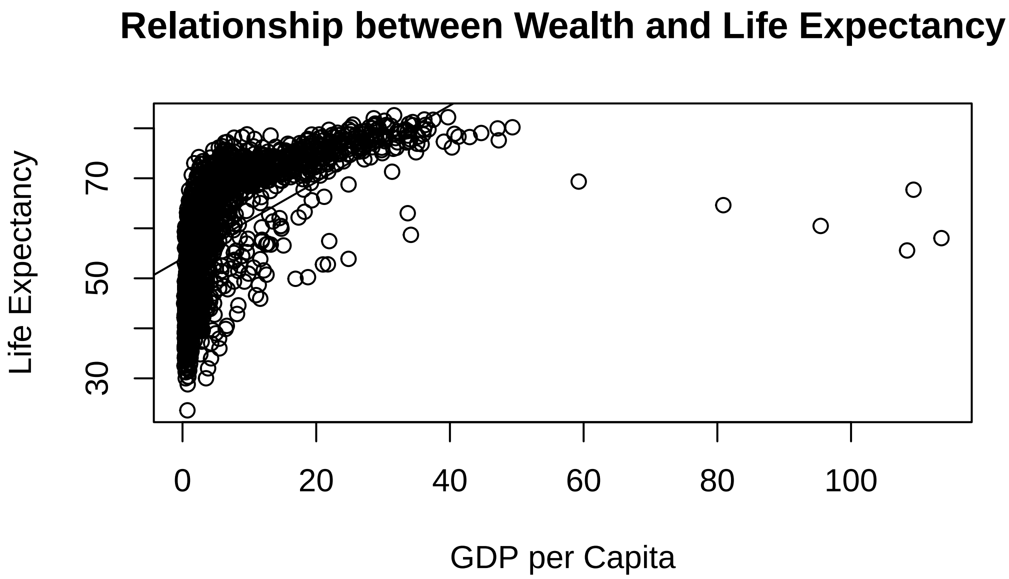
Notice how we always put the explanatory variable on the X axis(horizantal) and the outcome variable on the Y axis (vertical)

GDP per capita hard to read - change unit to thousands

```
gapminder$gdpPercap <- gapminder$gdpPercap/1000
```

Make plot look a bit more professional

```
plot(gapminder$gdpPercap, gapminder$lifeExp, main="Relationship between Wealth and Life Expec
abline(lm(gapminder$lifeExp~gapminder$gdpPercap)) #and boom, you've already ran your 2nd reg
```



Does there appear to be a relationship in the sample?

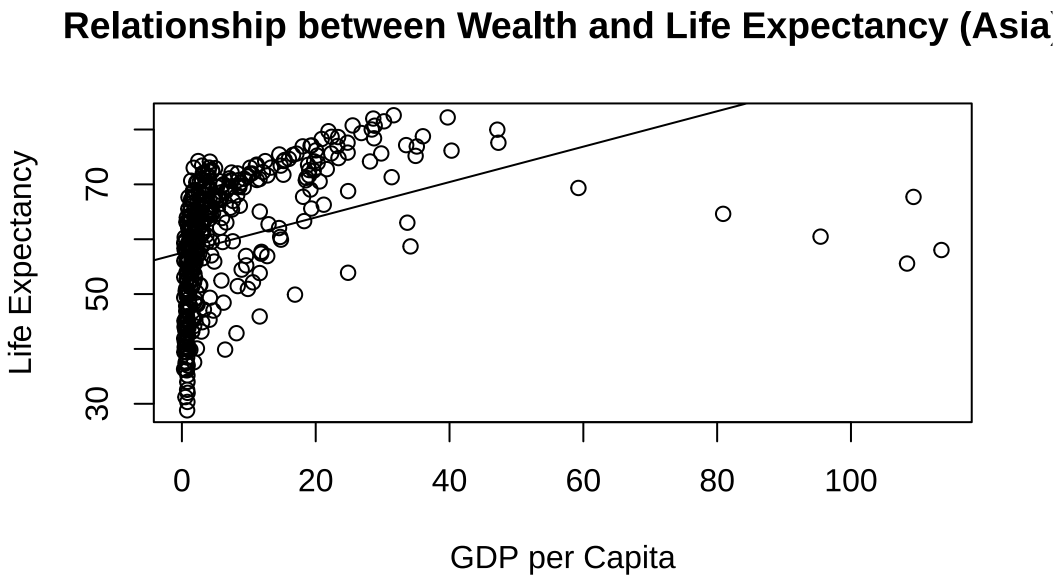
What about with just Asia? There are multiple ways to do this:

We can create a new dataset ie a subset(I almost always choose this option)

```
table(gapminder$continent)
```

```
  Africa Americas     Asia   Europe  Oceania 
     624      300      396      360       24 
```

```
asia <- gapminder[which(gapminder$continent=="Asia"),]
asia <- filter(gapminder, continent=="Asia") # This does the same thing using the dplyr packa



plot(asia$gdpPercap, asia$lifeExp, main="Relationship between Wealth and Life Expectancy (Asi
abline(lm(asia$lifeExp~asia$gdpPercap)) #and boom, another regression
```



Exclude Europe

```
noteuro <- gapminder[which(gapminder$continent!="Europe"),] # the != does not equal
noteuro <- filter(gapminder, continent!="Europe")
```

Exclude the super rich - use greater than or less than

```
notrich <- gapminder[which(gapminder$gdpPercap< 50),] # less than 50,000 - this drops oil-ri
notrich <- filter(gapminder, gdpPercap< 50)
```

Check in: Try creating a scatterplot with just the 1) Africa OR 2) country-years where life expectancy is less than 70.

#Looking at the data over time: Time to look at something a bit more fun. How has life expectancy changed across the world since the 1950s?

Mean life expectancy?

```
mean(gapminder$lifeExp)
```

```
[1] 59.47444
```

```
min(gapminder$lifeExp)
```

```
[1] 23.599
```

```r
max(gapminder$lifeExp)
```

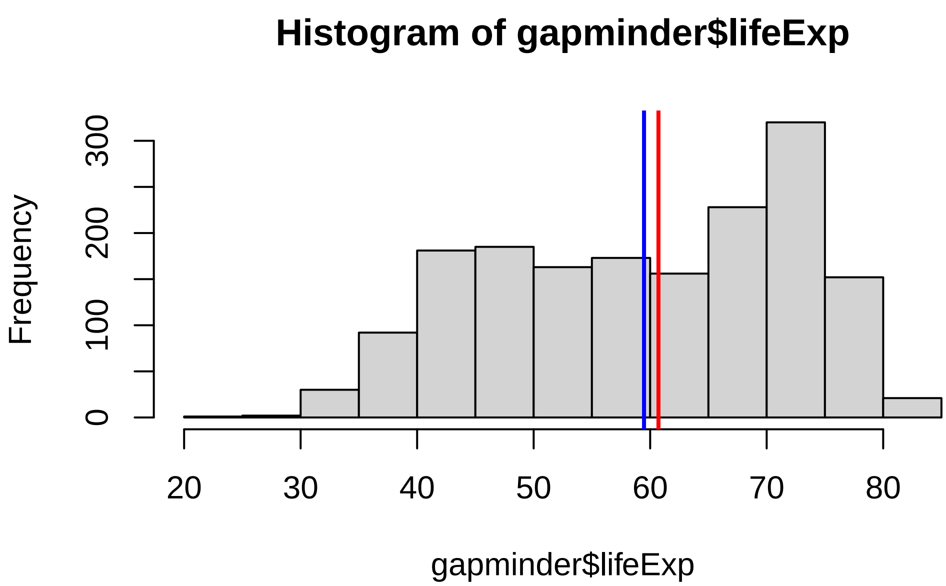
```
[1] 82.603
```

```r
range(gapminder$lifeExp)
```

```
[1] 23.599 82.603
```

Create a histogram for life expectancy

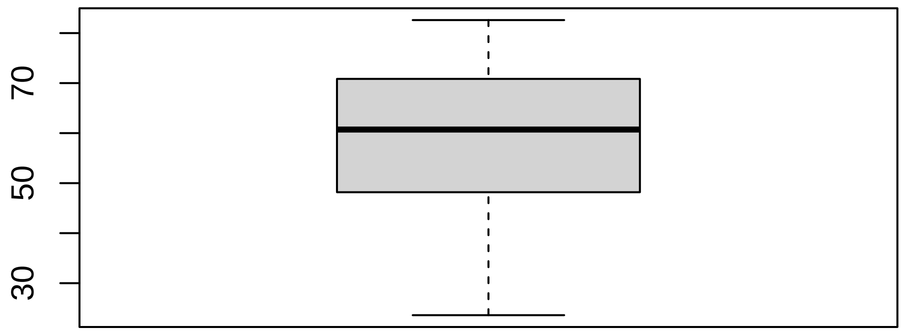
```r
hist(gapminder$lifeExp)
abline(v=mean(gapminder$lifeExp), col="blue", lwd=2)#adds a vertical(v) line for the mean
abline(v=median(gapminder$lifeExp), col="red", lwd=2)#a a line for mediant
```



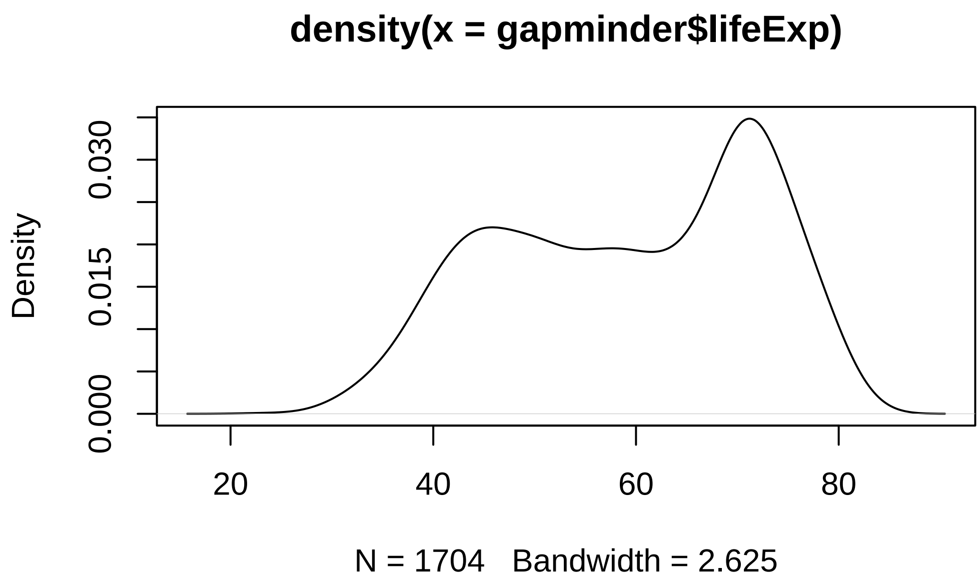
abline gives us a line. v is for vertical

Density plots and boxplots show us similar information

```r
boxplot(gapminder$lifeExp)
```




```
plot(density(gapminder$lifeExp))
```




```
#How can we find out which country-years are at the very bottom?
```

Check: How many observations are from Africa and how many are from Europe

```
mean(gapminder$lifeExp[gapminder$continent=="Africa"]) #average for Africa
```

```
[1] 48.86533
```

```
mean(gapminder$lifeExp[gapminder$continent=="Europe"]) #average for Europe
```

```
[1] 71.90369
```

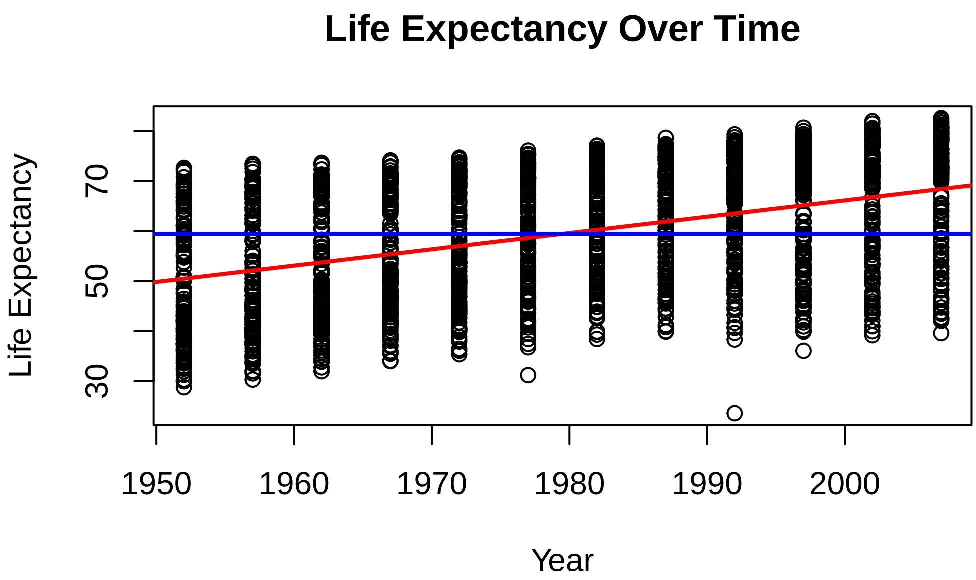
```
mean(asia$lifeExp)
```

```
[1] 60.0649
```

```
#find the mean for the Americas
```

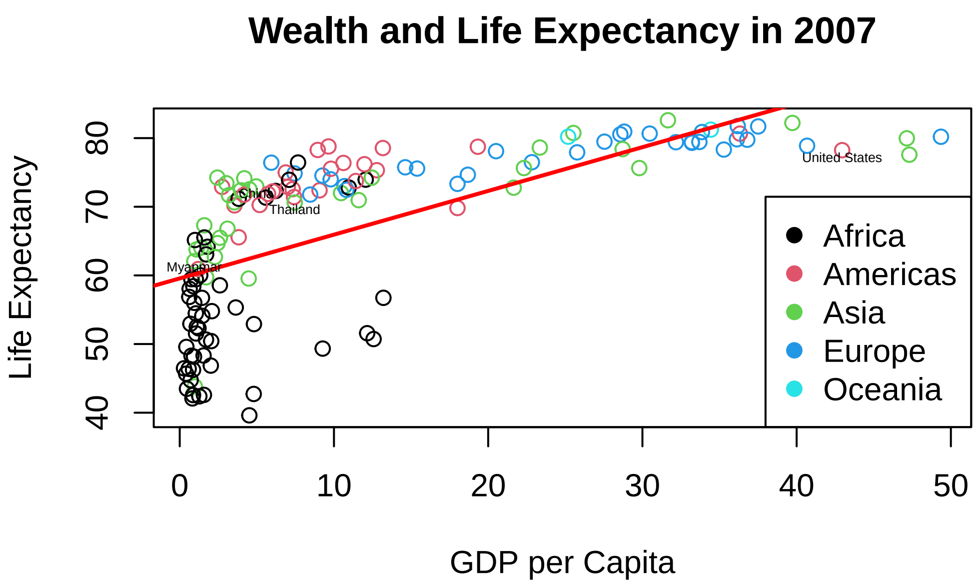
Let's make a simple scatterplot with year on the horizontal axis, and life expectancy on the vertical axis. A regression allows us to look at the relationship between variables linear regression model, and you're looking at the effect of the IV on the DV

```
plot(gapminder$year, gapminder$lifeExp, ylab = "Life Expectancy", xlab = "Year", main = "Life
abline(lm(gapminder$lifeExp ~ gapminder$year), col="red", lwd=2) # What type of relationship
abline(h=mean(gapminder$lifeExp), col="blue", lwd=2)
```



That is a bit overwhelming lets just look at 2007 - ie lets create a subset

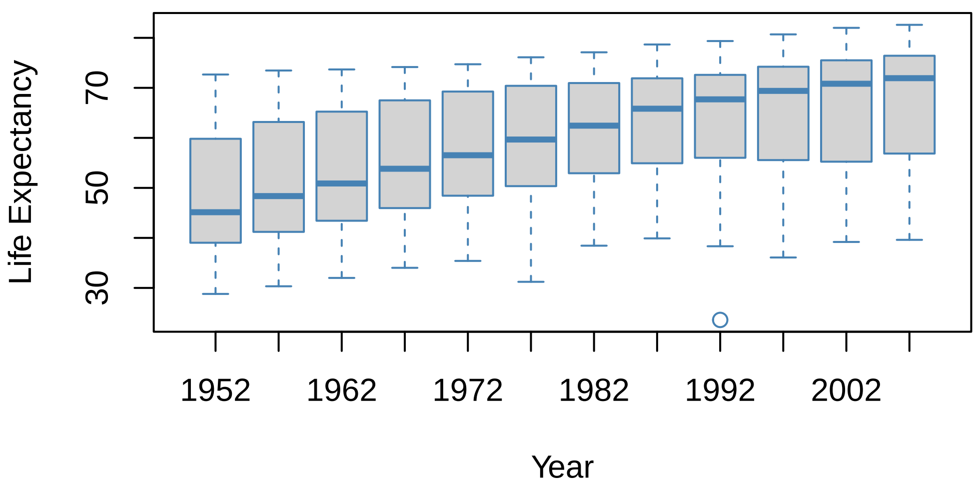
```
g2007 <- gapminder[which(gapminder$year==2007),] #just using base R

#New Graph
plot(g2007$gdpPercap, g2007$lifeExp, ylab = "Life Expectancy", xlab = "GDP per Capita", main
abline(lm(g2007$lifeExp ~ g2007$gdpPercap), col="red", lwd=2)
legend("bottomright", legend = levels(factor(g2007$continent)), pch=19, col = factor(levels(

#I want to add labels for specific countries
g2007$stars <- ifelse(g2007$country=="United States"|g2007$country=="China" |g2007$country=='
text(g2007$gdpPercap, g2007$lifeExp-1, g2007$stars, cex=.4) #add text to location
```



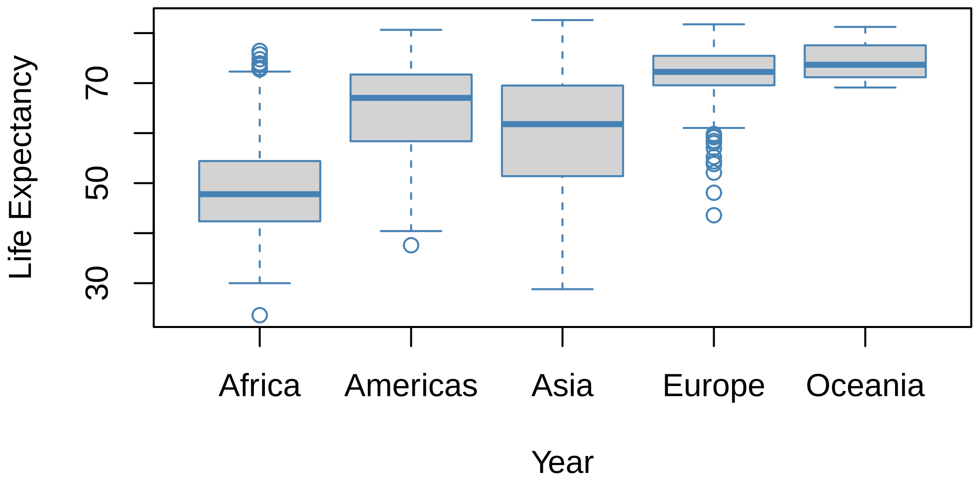
Check: Can you create the same for 1952

A boxplot for each year can show us how the distribution has changed over time

```
boxplot(gapminder$lifeExp~gapminder$year, xlab = "Year", ylab="Life Expectancy", border = "s
```

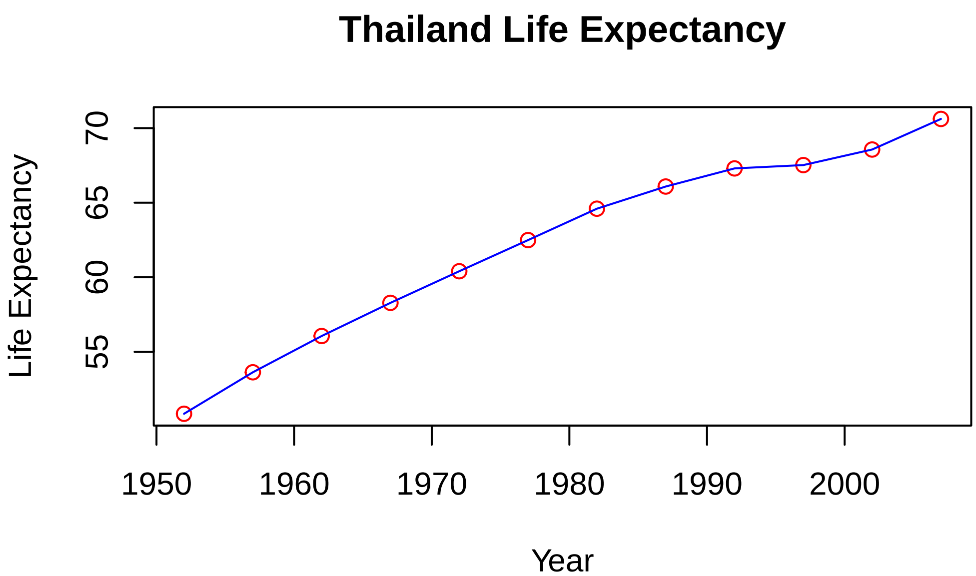



```
boxplot(gapminder$lifeExp~gapminder$continent, xlab = "Year", ylab="Life Expectancy", border
```

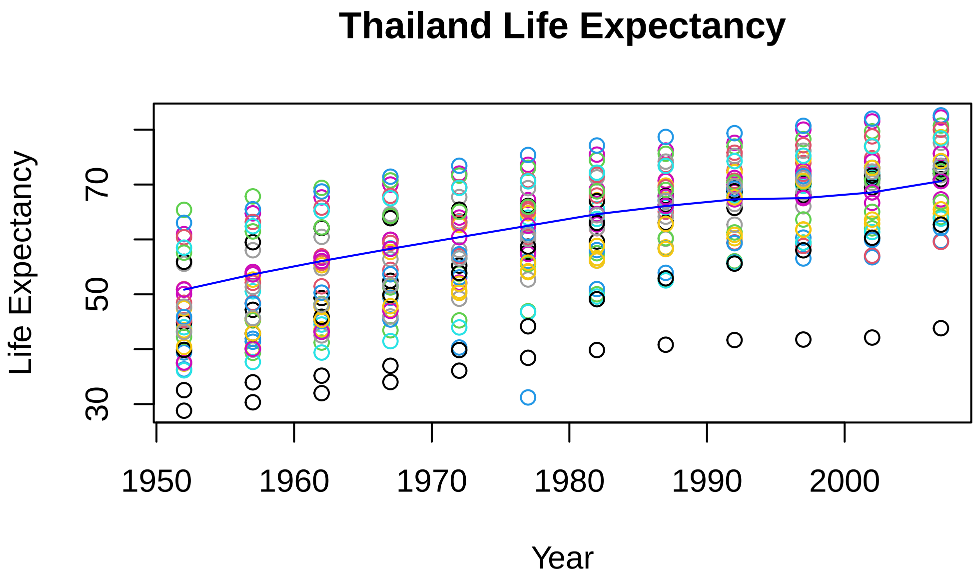


Lets look at changes in thailand over time

```
thai <- filter(gapminder, country=="Thailand")
plot(thai$year, thai$lifeExp, xlab="Year", ylab="Life Expectancy", col="red", main="Thailand
lines(thai$year, thai$lifeExp, col="blue")
```




```
plot(asia$year, asia$lifeExp, xlab="Year", ylab="Life Expectancy", col=gapminder$country, ma
lines(thai$year, thai$lifeExp, col="blue")
```



Check: Do the same above for a different country

I've clearly had too much fun. Your turn to make something interesting.

Since we've created some new variables we may want to save our data:

```
write.csv(g2007, "g2007.csv") #writes a csv to your working directory that you can open in e
save(g2007, file="g2007.Rdata") #saves Rdata format to your working directory
```